

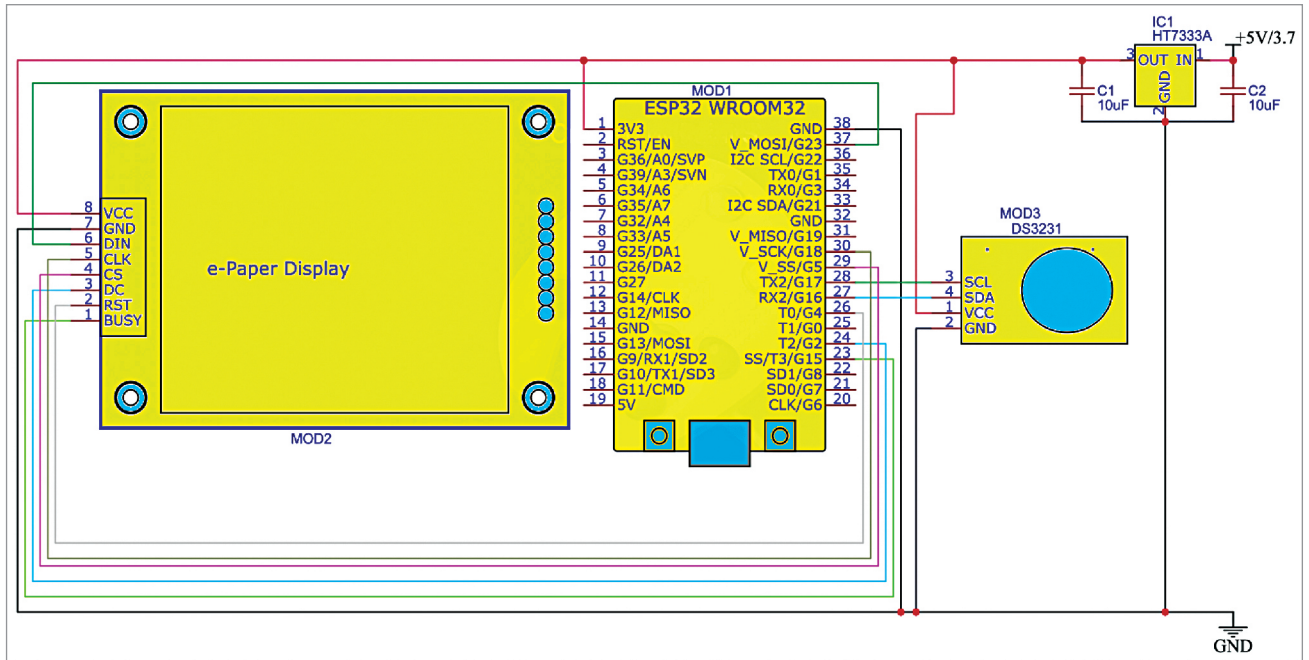


Fig. 2: Circuit diagram

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ESP32S2S Dev Module
aper_RTC_clock_with_calendar.ino  GxEPD2_display_selection.h  GxEPD2_selection_check.h  GxEPD2_wiring_examples.h
9
10
11 void thickRect(int x0,int y0, int x1, int y1, int thk,int clr, int color){
12   thickLine(x0,y0+clr+clr/2,x1-clr,y0+clr+clr/2,thk,color);
13   thickLine(x1-clr,y0+clr,x1-clr,y1-clr,thk,color);
14   thickLine(x1-clr,y1-clr,x0+clr,y1-clr,thk,color);
15   thickLine(x0+clr,y1-clr,x0+clr,y0+clr,thk,color);
16 }
17
18 void thickLine(int x, int y, int x1, int y1, int size, int color) {
19   float dx = (size / 2.0) * (y - y1) / sqrt(sq(x - x1) + sq(y - y1));
20   float dy = (size / 2.0) * (x - x1) / sqrt(sq(x - x1) + sq(y - y1));
21   display.fillTriangle(x + dx, y - dy, x - dx, y + dy, x1 + dx, y1 - dy, color);
22   display.fillTriangle(x - dx, y + dy, x1 - dx, y1 + dy, x1 + dx, y1 - dy, color);
23 }
24
25 void thickcircle(int16_t x0, int16_t y0, int16_t r, int pixel, uint16_t color) {
26   for(int i=r;i<=r+pixel;i++){
27     display.drawCircle(x0, y0, i, color);
28   }
29 }
30
31 void drawClockFace(){
32   display.fillCircle(ccenterx - hshift, ccentry-vshift, cradius-10, GxEPD_WHITE); //blue
33   display.fillCircle(ccenterx - hshift, ccentry-vshift, cradius-14, GxEPD_BLACK); //black
34   // Draw 12 lines
35   for(int i = 0; i<360; i+= 30) {
36     hr = i/30;
37     if(hr==0) hr=12;
38     // Serial.println(hr);
39     sx = cos((i-90)*scosConst);
40     sy = sin((i-90)*scosConst);
41     x0 = sx*(cradius-4-10)+ccenterx;
42     yy0 = sy*(cradius-4-10)+ccentry;
43     x1 = sx*(cradius-11-10)+ccenterx;
44     yy1 = sy*(cradius-11-10)+ccentry;
45     display.drawLine(x0, yy0-vshift, x1, yy1-vshift, GxEPD_WHITE); //blue
46     display.setTextColor(GxEPD_WHITE,GxEPD_BLACK);
47     int x00 = sx*(cradius-4-10-10)+ccenterx;
48     int yy00 = sy*(cradius-4-10-10)+ccentry;

```

Fig. 3: Code snippet

reverse colours will cancel each other and you will get your figures displayed properly.

In the first page, if you write in white colour on a black border, in the next page, you must write the same figure (or text) in black colour on white border. Failure to do so will result in a flickering ghost image that changes continuously. This is a trick we used to have the display change successfully every second inside the loop. Fig. 3 shows a code snippet of the source code.

Construction and testing

Assemble and connect all the components as per Fig. 2. For the connection HAT (marked on the backside of the display), ensure that the SPI connection switch is towards the '4-wire' SPI side, and the display configure switch is towards the 'B' side; otherwise, the display may not work! The ribbon cable between the e-paper and the HAT is very delicate. Try not to move this cable much, as it may permanently damage the e-paper. Next, connect the RTC module